People Analytics

Project Report

at the Faculty of Business, Economics, and Law
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Content

Content 2

1 Well-Being 3
1.1 The importance of well-being *(4 Points)* 3
1.2 Stress induced by technology *(7 Points)* 4
1.3 Employee well-being dataset *(8 Points)* 4
1.4 Predictive well-being algorithm *(6 Points)* 8

2 Turnover 13
2.1 Employee Turnover *(4 Points)* 13
2.2 Employee Turnover Theories *(6 Points)* 14
2.3 Employee Turnover Dataset *(10 Points)* 15
2.4 Turnover prediction *(5 Points)* 20

References VII

Declaration of Academic Integrity at the Schöller Endowed Chair for Information Systems VIII

1 Well-Being

1.1 The importance of well-being *(4 Points)*

Employee well-being plays a central role in maintaining a healthy, productive, and engaged workforce at FAU Transline. Monitoring employee well-being is not merely a moral obligation but also a strategic imperative that directly impacts organizational performance. According to Danna and Griffin (1999), well-being in the workplace encompasses both job-related satisfaction and overall life satisfaction, influencing both physical and psychological health. Organizations that actively monitor and improve well-being often experience lower absenteeism, reduced healthcare costs, and increased employee morale and productivity. If well-being is ignored, it can lead to increased stress, burnout, job dissatisfaction, and ultimately, higher turnover—especially relevant as FAU Transline plans to expand operations and staffing. A proactive focus on well-being aligns with the company's goal of creating a supportive and efficient work environment.

Factors Leading to Low Well-Being and Poor Work-Life Balance

Several factors can contribute to reduced well-being and an unhealthy work-life balance among employees at FAU Transline:

1. **Job Design and Autonomy**: Lack of control over tasks and rigid job structures contribute to stress and lower engagement.
2. **Long Working Hours**: Extensive shifts, especially for conductors and drivers, are linked to fatigue and chronic stress.
3. **Technostress**: Increased use of ICT systems (like the planned software replacement) can cause techno-overload, techno-insecurity, and other stressors
4. **Poor Social Support**: Limited interaction with colleagues or supervisors reduces emotional resilience and social well-being.
5. **Perceived Unfairness**: Inequitable treatment, such as favoritism or lack of recognition, can diminish morale.
6. **Work-Family Conflict**: Irregular schedules can interfere with family responsibilities, increasing stress and dissatisfaction.

FAU Transline can monitor well-being by conducting regular employee surveys, tracking key metrics like absenteeism and stress levels, and using data analytics to identify trends. Managers should hold routine check-ins, and results should guide targeted wellness programs.

1.2 Stress induced by technology *(7 Points)*

Replacing FAU Transline's legacy software system (ROMS) with a modern digital platform may unintentionally increase **technostress** among employees. Technostress arises when ICTs (Information and Communication Technologies) introduce challenges beyond what employees can comfortably handle.

Potential Stressors and Consequences:

1. **Techno-Overload:** New digital systems may require employees to complete more tasks in less time, increasing the workload.
2. **Techno-Complexity**: Employees unfamiliar with the new system may feel overwhelmed, lacking the skills to adapt.
3. **Techno-Insecurity**: Fear of being replaced by technology or being seen as incompetent may reduce morale.
4. **Techno-Uncertainty**: Constant updates and system changes can lead to frustration and mental fatigue

These stressors can lead to negative outcomes, including reduced job satisfaction, increased burnout, lower performance, and eventually turnover.

Coping Strategies and Recommendations:

1. **Training & Support**: Offer hands-on training and continuous support to reduce fear and boost confidence.
2. **Inclusive Implementation**: Involve employees early in the transition process to foster a sense of control and reduce resistance.
3. **Gradual Rollout**: Implement the new system in phases to allow adaptation and reduce shock.

By proactively addressing these challenges, FAU Transline can reduce technostress and ensure a smoother digital transformation aligned with employee well-being.

1.3 Employee well-being dataset *(8 Points)*

Objective:

The goal of this analysis is to identify the main factors that influence the **WORK_LIFE_BALANCE_SCORE (WLB)** of employees at FAU Transline. The dataset ***wellbeing__of_employees.csv*** was used for this purpose. The following steps were followed during the analysis:

1. Preprocessing the data by handling missing values, selecting relevant features, and transforming categorical data.

2. Visualizing daily stress based on gender and age to understand demographic stress patterns.

3. Identifying which gender takes more daily steps using grouped comparisons.

4. Generating a correlation matrix to find variables most strongly associated with the WLB score.

Step 1: Data Preprocessing

1 Checking for Missing Values

- Missing values were found in the *DAILY_STRESS* column due to invalid entries.
- These were coerced to numeric format, and invalid rows were dropped.

2 Feature Selection

- **Dropped Column:** *BMI_RANGE*
  - Reason 1: Very low correlation with WLB Score.
  - Reason 2: No direct theoretical relevance to work-life balance.

3 Converting String Datatypes to Numeric

- **AGE:**
  - "Under 20" → 0
  - "21 to 35" → 1
  - "36 to 50" → 2
  - "51 or more" → 3
- **GENDER:**
  - "Female" → 0
  - "Male" → 1

Code Snippet:

```
# Data Preprocessing
data['DAILY_STRESS'] = pd.to_numeric(data['DAILY_STRESS'], errors='coerce')
data.dropna(subset=['DAILY_STRESS'], inplace=True)
data.drop(columns=['BMI_RANGE'], inplace=True)

# Mapping categorical to numeric
age_mapping = {"Under 20": 0, "21 to 35": 1, "36 to 50": 2, "51 or more": 3}
gender_mapping = {"Female": 0, "Male": 1}
data['AGE'] = data['AGE'].map(age_mapping)
data['GENDER'] = data['GENDER'].map(gender_mapping)
```

Step 2: Visualizing Insights

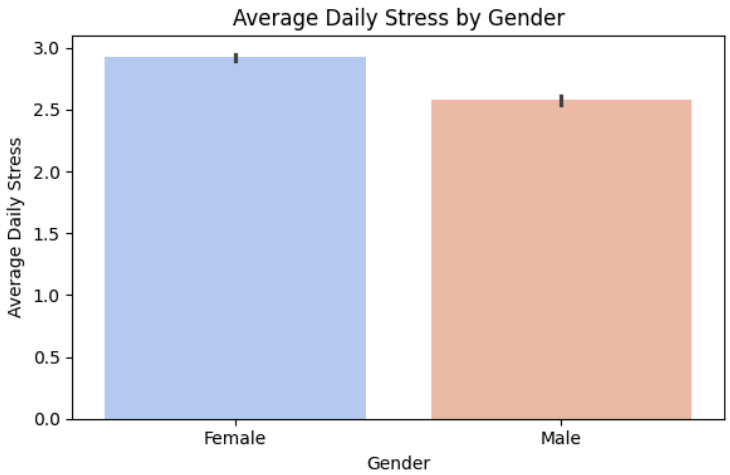
1 Daily Stress by Gender

- **Observation:**

  Females report slightly higher average daily stress than males.
- **Code Snippet:**

```
# Plot: Average Daily Stress by Gender
plt.figure(figsize=(6, 4))
sns.barplot(x='GENDER', y='DAILY_STRESS', data=data, palette='coolwarm')
plt.title('Average Daily Stress by Gender')
plt.xticks([0, 1], ['Female', 'Male'])
plt.xlabel('Gender')
plt.ylabel('Average Daily Stress')
plt.tight_layout()
plt.savefig("Average Daily Stress by Gender.png")
plt.show()
```

- **Visualization:**



2 Daily Stress by Age

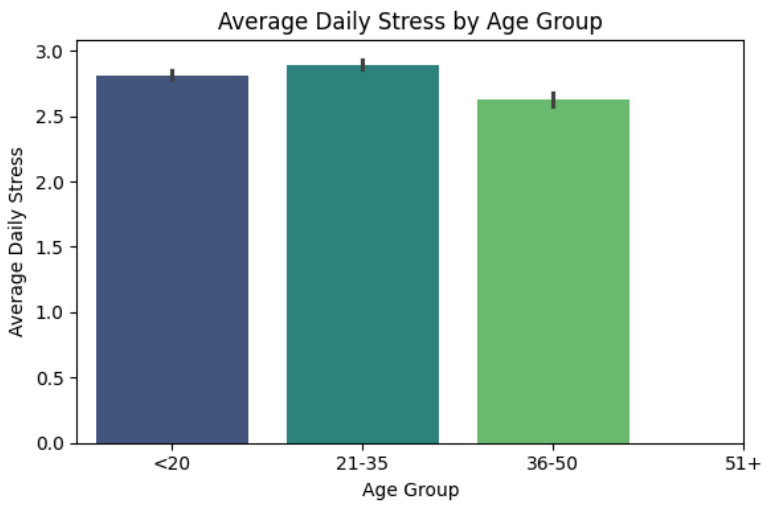
- **Observation:**

  Stress peaks in employees aged “21–35”, then slightly decreases.

- **Code Snippet:**

```
# Plot: Average Daily Stress by Age Group
plt.figure(figsize=(6, 4))
sns.barplot(x='AGE', y='DAILY_STRESS', data=data, palette='viridis')
plt.title('Average Daily Stress by Age Group')
plt.xticks(ticks=[0, 1, 2, 3], labels=["<20", "21-35", "36-50", "51+"])
plt.xlabel('Age Group')
plt.ylabel('Average Daily Stress')
plt.tight_layout()
plt.savefig("Average Daily Stress by Age Group.png")
plt.show()
```

- **Visualization:**



3 Daily Steps by Gender

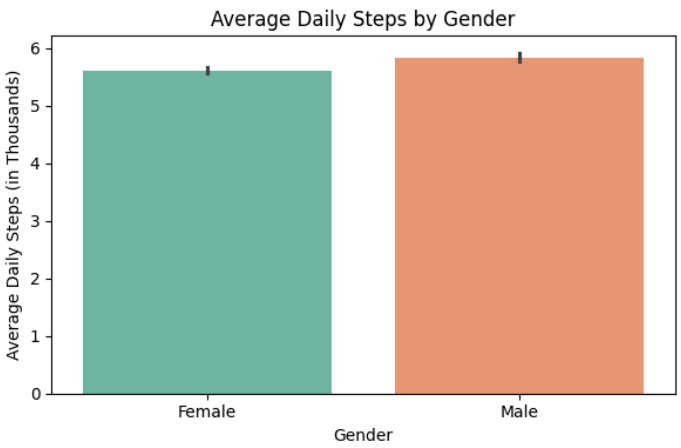
- **Observation:**

  Males take slightly more daily steps than females.

- **Code Snippet:**

```
# Plot: Daily Steps by Gender
plt.figure(figsize=(6, 4))
sns.barplot(x='GENDER', y='DAILY_STEPS_IN_THOUSAND', data=data, palette='Set2')
plt.title('Average Daily Steps by Gender')
plt.xticks([0, 1], ['Female', 'Male'])
plt.xlabel('Gender')
plt.ylabel('Average Daily Steps (in Thousands)')
plt.tight_layout()
plt.savefig("Daily Steps by Gender.png")
plt.show()
```

- **Visualization:**



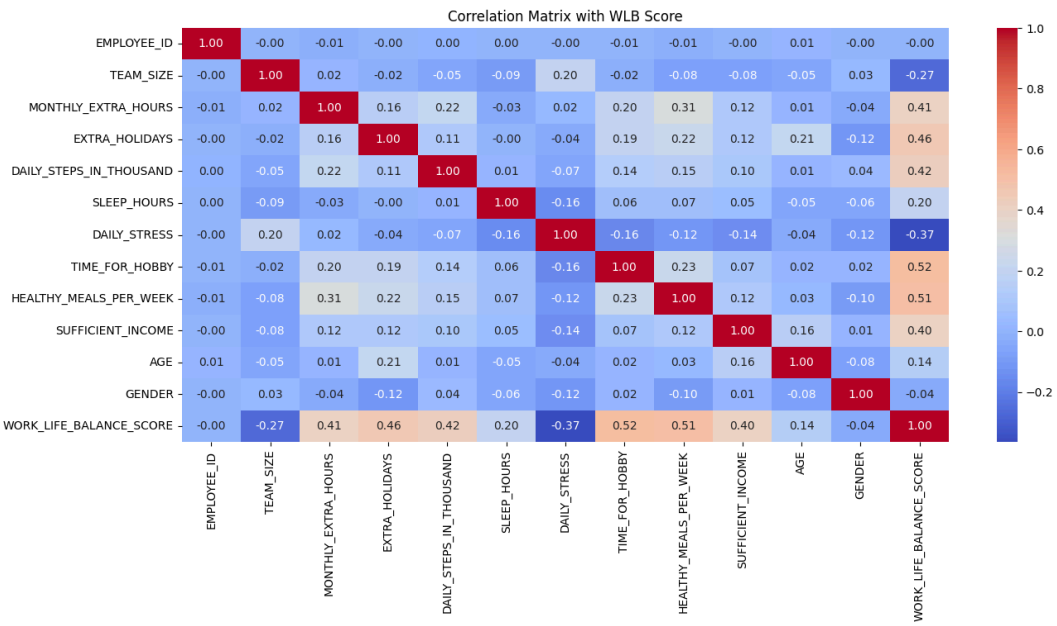
Step 3: Correlation Analysis

- **Key Correlations with WLB:**
  - Positive Correlations:
    - TIME_FOR_HOBBY → (0.52)
    - HEALTHY_MEALS_PER_WEEK → (0.51)
    - EXTRA_HOLIDAYS → (0.46)
    - DAILY_STEPS_IN_THOUSAND → (0.42)
    - MONTHLY_EXTRA_HOURS → (0.41)
    - SUFFICIENT_INCOME → (0.40)
  - Negative Correlation:
    - DAILY_STRESS → (- 0.37)

- **Code Snippet:**

```
# Plot: Correlation Matrix with WLB Score
plt.figure(figsize=(10, 8))
correlation_matrix = data.corr(numeric_only=True)
sns.heatmap(correlation_matrix, annot=True, fmt=".2f", cmap='coolwarm', cbar=True)
plt.title('Correlation Matrix with WLB Score')
plt.tight_layout()
plt.savefig("Correlation Matrix with WLB Score.png")
plt.show()
```

- **Visualization:**



1.4 Predictive well-being algorithm *(6 Points)*

1. Model Evaluation:

We used a **Linear Regression** model to predict the **WORK_LIFE_BALANCE_SCORE (WLB)** for employees based on features like gender, daily stress, hobbies, sleep hours, and others.

Code Snippet:

```
# Step 5: Modeling pipeline
model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("regressor", LinearRegression())
])
# Step 6: Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
# Step 7: Fit the model
model.fit(X_train, y_train)
```

R^2 Score:

```
# Step 8: Predict and evaluate
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
print(f"R² Score: {r2:.4f}")
```

Output:

```
R² Score: 0.8592
```

- **R^2 Score:** `0.8592`
- This means that approximately **85.92%** of the variance in the WLB scores can be explained by the model.

2. Real vs. Predicted Values:

We present the difference between the **real WLB scores** and the **predicted WLB scores** in tabular format.

Code Snippet:

```
# Step 9: Show real vs predicted in table
comparison_df = pd.DataFrame({
    "Real WLB": y_test.values,
    "Predicted WLB": np.round(y_pred, 2),
    "Difference": np.round(y_test.values - y_pred, 2)
}).reset_index(drop=True)
```

Output:

```
Real vs Predicted (Sample):
   Real WLB  Predicted WLB  Difference
0     655.4         643.09       12.31
1     630.8         624.01        6.79
2     694.0         719.23      -25.23
3     638.0         637.34        0.66
4     647.1         629.95       17.15
```

The Difference column reflects the prediction error. Most predictions are accurate, with some discrepancies.

3. Prediction for New Employee:

We predicted the WLB score for a hypothetical employee with the following characteristics:

- **Job Role**: Train Conductor
- **Team Size**: 6
- **Monthly Extra Hours**: 5
- **Extra Holidays**: 3
- **To-Dos Completed**: 7
- **BMI Range**: 1
- **Daily Steps**: 6
- **Sleep Hours**: 7
- **Daily Stress**: 3
- **Time for Hobby**: 4
- **Healthy Meals per Week**: 7
- **Sufficient Income**: 2
- **Age**: 21 to 35
- **Gender**: Male

Code Snippet:

```
# Step 10: Predict new employee
new_employee = pd.DataFrame([{
    'JOB_ROLE': 'Train Conductor',
    'TEAM_SIZE': 6,
    'MONTHLY_EXTRA_HOURS': 5,
    'EXTRA_HOLIDAYS': 3,
    'TODO_COMPLETED': 7,
    'BMI_RANGE': 1,
    'DAILY_STEPS_IN_THOUSAND': 6,
    'SLEEP_HOURS': 7,
    'DAILY_STRESS': 3,
    'TIME_FOR_HOBBY': 4,
    'HEALTHY_MEALS_PER_WEEK': 7,
    'SUFFICIENT_INCOME': 2,
    'AGE': '21 to 35',
    'GENDER': 'Male'
}])
```

Output:

```
Predicted WLB Score for New Employee: 683.35
```

The **predicted WLB score** for this employee is **683.35**.

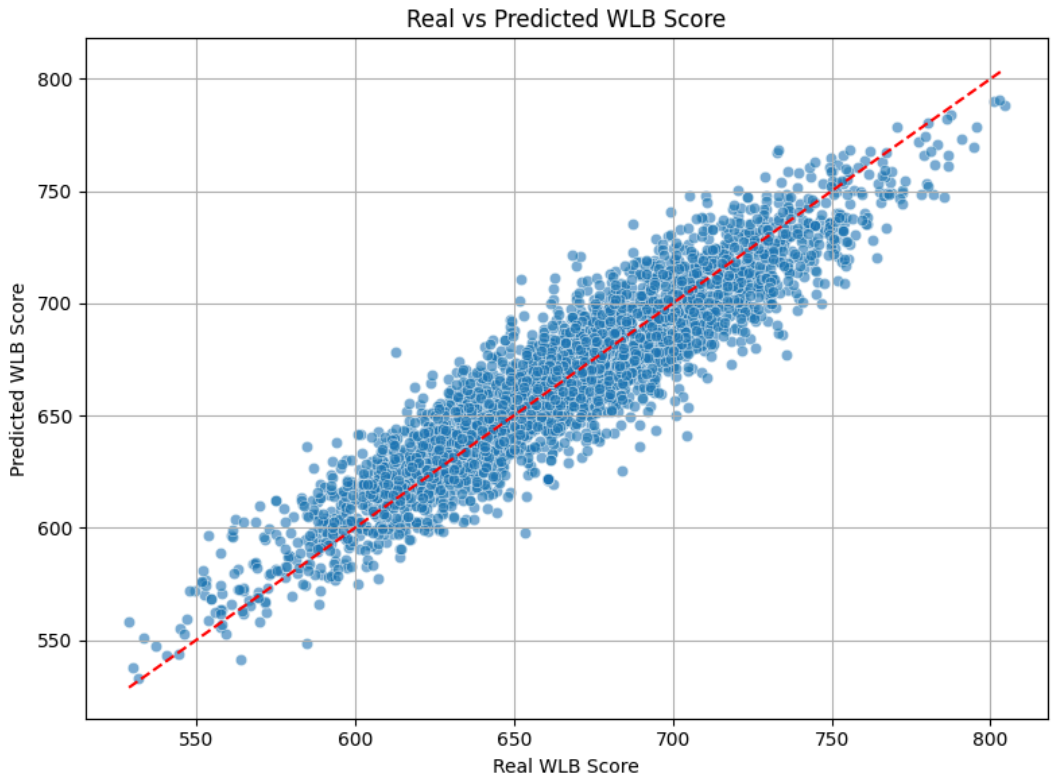
4. Visual Representation:

We plotted a **scatter plot** comparing real and predicted WLB scores:

Code Snippet:

```
# Step 11: Scatter plot: Real vs Predicted
plt.figure(figsize=(8, 6))
sns.scatterplot(x=y_test, y=y_pred, alpha=0.6)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--')
plt.xlabel('Real WLB Score')
plt.ylabel('Predicted WLB Score')
plt.title('Real vs Predicted WLB Score')
plt.grid(True)
plt.tight_layout()
plt.savefig("Real vs Predicted WLB Score.png")
plt.show()
```

Output:



This plot shows that most predictions align closely with actual values along the line $y = x$.

5. Recommendations for FAU Transline:

Based on the model and predictions, the following recommendations are made:

1. **Promote Physical Activity**

   Encourage walking and healthy routines—higher step counts relate to better WLB.

2. **Offer More Leisure Flexibility**

   Employees with more hobby time show better balance—promote hobby hours or flexible scheduling.

3. **Monitor Overtime**

   Limit monthly extra hours to reduce stress and improve WLB scores.

4. **Boost Healthy Eating**

   Provide access to healthy meals—more healthy meals per week improve well-being.

5. **Allow Extra Holidays**

   Granting 2–5 extra holidays could enhance morale and balance.

2 Turnover

2.1 Employee Turnover *(4 Points)*

Voluntary employee turnover refers to situations where employees choose to leave the organization of their own accord, rather than being terminated or laid off. This type of turnover typically occurs when employees are dissatisfied with their job, seek better opportunities, or feel disconnected from the organization.

Why It Is a High Concern for FAU Transline

Voluntary turnover is a major concern for FAU Transline for several reasons:

- **Operational Disruption:** Frequent turnover among conductors, drivers, and technicians disrupts public transportation schedules and customer service reliability.
- **Training and Recruitment Costs:** Replacing employees involves recruitment, onboarding, and training costs, which are time-consuming and expensive.
- **Loss of Experienced Talent:** Employees who leave often take their experience and expertise with them, which reduces institutional knowledge.
- **Team Morale and Productivity:** High turnover can demotivate remaining employees, increase workloads, and reduce morale.

Given FAU Transline's plans for expansion and its need for a stable workforce to operate new routes and hubs, retaining talent is crucial for consistent service delivery and organizational growth.

Organizational Equilibrium Theory and Causes of Turnover

The Organizational Equilibrium Theory, proposed by March and Simon (1958), suggests that an employee's decision to stay or leave depends on the balance between the inducements (rewards, satisfaction, recognition) offered by the organization and the contributions (effort, time, skills) expected from the employee.

According to this theory, voluntary turnover occurs when:

- The **inducements fall below expectations** (e.g., insufficient compensation, low job satisfaction, lack of recognition).
- The **required contributions increase disproportionately** (e.g., more workload without corresponding benefits).
- **Alternative opportunities outside** the organization become more attractive than staying.

In the context of FAU Transline, employees may leave if:

- The job does not meet their financial or emotional needs (e.g., insufficient income or work-life balance).
- There is a lack of growth or career development.
- Changes in work structure (such as centralization or standardization) reduce autonomy and employee engagement.

Recommendation:
To reduce voluntary turnover, FAU Transline should:

- Improve job satisfaction through regular feedback, recognition, and career development programs.
- Ensure that compensation aligns with market standards and workload.
- Foster a sense of belonging by promoting team cohesion and flexible work environments.

2.2 Employee Turnover Theories *(6 Points)*

Case 1: Jonas – Person-Organization Fit Theory

Theory Explanation:

- Employees prefer workplaces where their **values and work styles** align with the organization's culture.
- A **mismatch** leads to dissatisfaction, disengagement, and voluntary turnover.

Application to Jonas:

1. Jonas values **independence, flexibility**, and **close team relationships**.
2. The company restructured with **centralized control, rigid procedures,** and **formalized operations.**
3. This caused a **misalignment** between his preferences and the new organizational structure.
4. The change led to a **loss of autonomy**, which clashed with his values.
5. This **poor fit** triggered his decision to resign.

Case 2: Elena – Belongingness Theory (Social Identity Theory)

Theory Explanation:

- Humans have a **psychological need to belong** and feel connected to a group.
- Lack of **social connection** leads to **emotional withdrawal** and increased risk of quitting.
- Belonging is vital for **job satisfaction** and **employee retention**.

Application to Elena:

1. Elena values the company's **customer-first mission** and enjoys her role.
2. She lacks **social interaction** and hasn't formed **strong workplace relationships**.
3. This creates **feelings of isolation** and emotional disconnection.
4. The **absence of social bonds** undermines her motivation to stay.
5. This lack of belonging is causing her to **consider leaving the company**.

Retention Strategy for Elena and Similar Cases:

To prevent such turnover, FAU Transline could:

- **Foster Team-Building Activities:** Create opportunities for conductors to bond outside of work (e.g., workshops, retreats, team lunches).
- **Mentorship Programs:** Pair new employees with more experienced colleagues to ease social integration.
- **Peer Support Groups:** Encourage the formation of informal groups based on shared interests or challenges.
- **Frequent Check-ins:** HR or team leads should regularly check in to understand social well-being, not just performance.

These initiatives can help employees like Elena feel more connected, ultimately improving their sense of belonging and reducing the risk of voluntary turnover.

2.3 Employee Turnover Dataset *(10 Points)*

The ***"turnover.csv"*** dataset contains employee-level data from FAU Transline, including attributes such as job satisfaction, salary, tenure, job role, and whether the employee has left the company (***`left = 1`***) or stayed (***`left = 0`***). This section aims to investigate key indicators behind employee turnover and perform data preprocessing to prepare the dataset for future predictive analysis.

1. Salary by Job Role

Analysis:

To assess salary variation, we used a grouped bar chart to show the count of employees at each **salary level** (*low, medium, high*) for each **job role**.

Code Snippet:

```
plt.figure(figsize=(12, 6))
sns.countplot(data=df_viz, x='job_role', hue='salary', palette='coolwarm', order=sorted(df_viz['job_role'].unique()))
plt.title('Salary by Job Role')
plt.xlabel('Job Role')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("Salary by Job Role.png")
plt.show()
```

Output:

The resulting bar chart reveals that **most job roles** are concentrated in the **low and medium salary levels**, with fewer employees receiving high salaries.

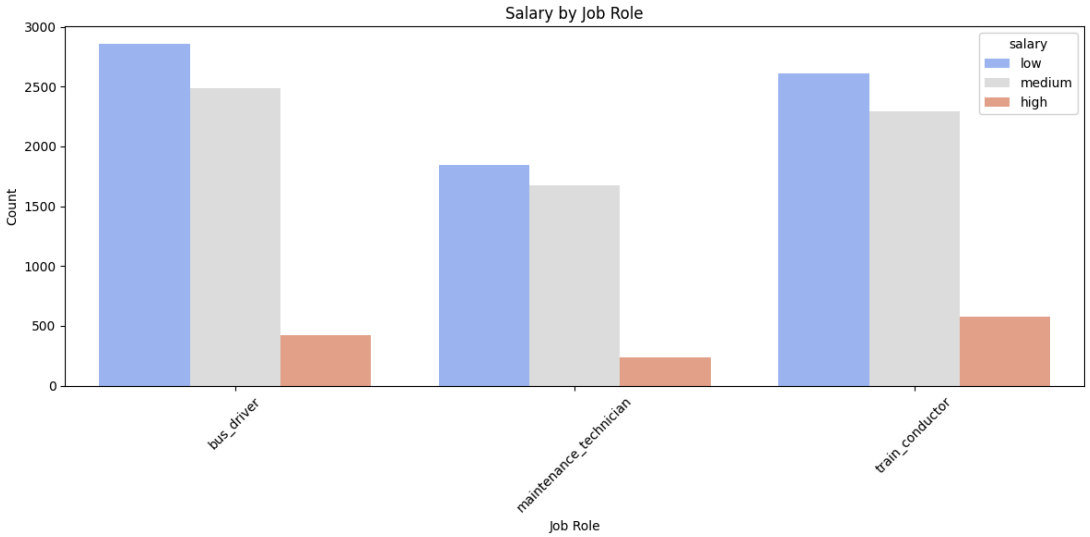


Figure 1: Salary by Job Role

2. Average Job Satisfaction of Employees Who Left

Analysis:

We computed the average satisfaction level of employees who voluntarily left the company.

Code Snippet:

```
avg_satisfaction_left = df[df['left'] == 1]['job_satisfaction_rate'].mean()
print(f"Average Job Satisfaction of Employees Who Left: {avg_satisfaction_left:.2f}")
```

Output:

```
Average Job Satisfaction of Employees Who Left: 0.44
```

This indicates a generally **low satisfaction** score among those who exited.

3. Tenure of Employees Who Left

Analysis:

We measured the average number of years employees stayed before leaving.

Code Snippet:

```
avg_time_spent_left = df[df['left'] == 1]['time_spend_company'].mean()
print(f"Average Time Spent by Employees Who Left: {avg_time_spent_left:.2f} years")
```

Output:

```
Average Time Spent by Employees Who Left: 3.88 years
```

Employees typically left after spending nearly **4 years** at the company.

4. Correlation Analysis with left

Analysis:

To understand the factors influencing employee turnover, we computed the **correlation matrix** and visualized it using a heatmap.

Code Snippet:

```
correlation_matrix = df.corr(numeric_only=True)
print("\nCorrelation with 'left':\n", correlation_matrix['left'].sort_values(ascending=False))

plt.figure(figsize=(10, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix')
plt.tight_layout()
plt.savefig("Correlation Matrix.png")
plt.show()
```

Output:

```
Correlation with 'left':
 left                        1.000000
time_spend_company          0.144822
average_monthly_hours       0.071287
number_colleague_friends    0.023787
job_role                   -0.016982
promotion_last_3years      -0.061788
job_role_market_demand     -0.118430
salary                     -0.157898
job_satisfaction_rate      -0.388375
Name: left, dtype: float64
```

- Most negatively correlated feature: `job_satisfaction_rate = -0.39`
- Others: `salary = -0.16`, `promotion_last_3years = -0.06`, `time_spend_company = +0.14`

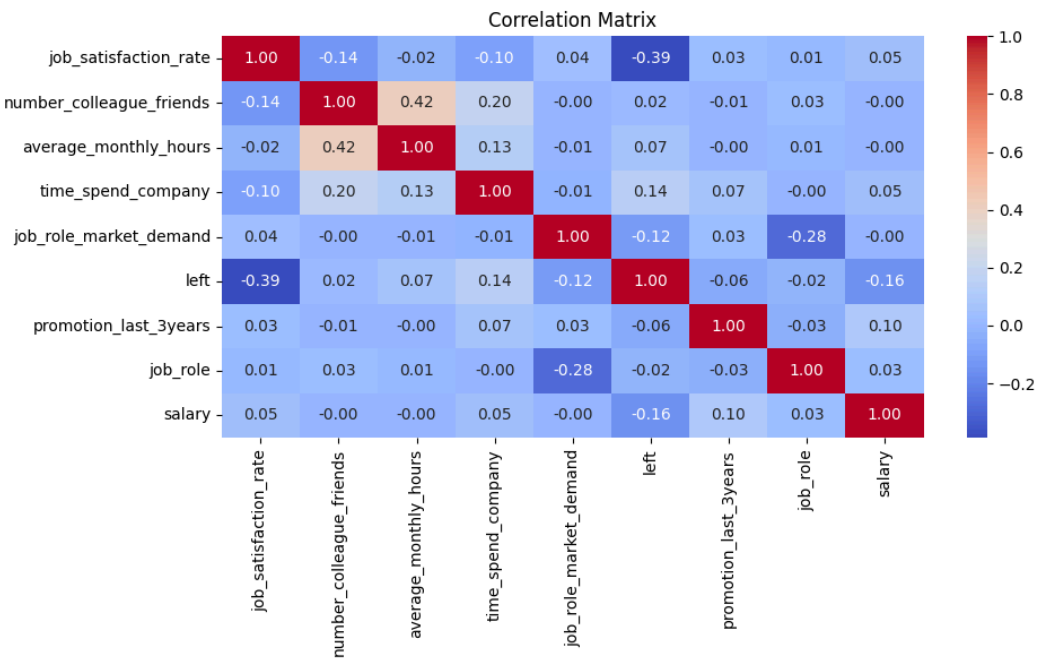


Figure 2: Correlation Matrix

5. Missing Values in the Dataset

Analysis:

We verified that the dataset was complete and clean.

Code Snippet:

```
missing_values = df.isnull().sum()
print("\nMissing Values in Dataset:")
print(missing_values)
```

Output:

```
Missing Values in Dataset:
job_satisfaction_rate       0
number_colleague_friends    0
average_monthly_hours       0
time_spend_company          0
job_role_market_demand      0
left                        0
promotion_last_3years       0
job_role                    0
salary                      0
dtype: int64
```

All columns have **0 missing values**.

6. Data Preprocessing

Step 1: Convert Categorical to Numeric

We mapped the following:

- `salary` → low = 0, medium = 1, high = 2
- `job_role` → Mapped using a dictionary based on unique values

Code Snippet:

```
df['salary'] = df['salary'].map({'low': 0, 'medium': 1, 'high': 2})
job_role_mapping = {role: idx for idx, role in enumerate(df['job_role'].unique())}
reverse_job_role_mapping = {v: k for k, v in job_role_mapping.items()}
df['job_role'] = df['job_role'].map(job_role_mapping)
```

Step 2: Final Preprocessed Dataset Sample

The first few rows of the transformed dataset confirm that both `salary` and `job_role` have been successfully mapped to numeric values.

Code Snippet:

```
print("\nPreprocessed Dataset Sample:")
print(df.head())
```

Output:

```
Preprocessed Dataset Sample:
   job_satisfaction_rate  number_colleague_friends  average_monthly_hours  time_spend_company  ...  left  promotion_last_3years  job_role  salary
0                   0.38                         2                    157                   3  ...     1                      0         0       0
1                   0.80                         5                    262                   6  ...     1                      0         0       1
2                   0.11                         7                    272                   4  ...     1                      0         0       1
3                   0.72                         5                    223                   5  ...     1                      0         0       0
4                   0.37                         2                    159                   3  ...     1                      0         0       0
```

This output preview shows a clean, numeric dataset ready for further statistical analysis and machine learning.

2.4 Turnover prediction *(5 Points)*

The goal is to develop a machine learning model that predicts employee turnover at **FAU Transline** using the provided dataset. Based on the prediction, we evaluate the factors contributing to turnover and suggest practical steps the organization can take to retain employees.

Model Selection

For predicting employee turnover, I selected the **Random Forest Classifier**. This model is known for:

- Handling both numerical and categorical data,
- Robustness to overfitting,
- Interpretability through feature importance rankings.

It was therefore an excellent choice for this classification task.

Data Preprocessing

We performed the following preprocessing steps:

1. **Label Encoding**:
   Converted categorical features `'job_role'` and `'salary'` into numerical form.
2. **Train-Test Split**:
   The dataset was split into **80% training** and **20% testing** sets.

Code Snippet – Preprocessing

```
# Copy the data
df_encoded = df.copy()

# Encode 'job_role' and 'salary'
df_encoded['job_role'] = LabelEncoder().fit_transform(df_encoded['job_role'])
df_encoded['salary'] = df_encoded['salary'].map({'low': 0, 'medium': 1, 'high': 2})

# Define features and target
X = df_encoded.drop('left', axis=1)
y = df_encoded['left']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Model Training

We trained a **Random Forest Classifier** using the training data. This ensemble model captures complex, non-linear relationships and provides feature importance metrics.

Code Snippet – Model Training

```
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Model Evaluation

We evaluated the model's performance using:

- Accuracy
- Precision
- Recall
- F1-score

Code Snippet – Evaluation

```
# Evaluation metrics
accuracy = accuracy_score(y_test, y_pred)
report = classification_report(y_test, y_pred, output_dict=True)
conf_matrix = confusion_matrix(y_test, y_pred)

# Print results
print(f"Accuracy: {accuracy}")
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Model Performance Output:

```
Accuracy: 0.9866666666666667
Classification Report:
               precision    recall  f1-score   support

           0       0.99      0.99      0.99      2294
           1       0.98      0.96      0.97       706

    accuracy                           0.99      3000
   macro avg       0.99      0.98      0.98      3000
weighted avg       0.99      0.99      0.99      3000
```

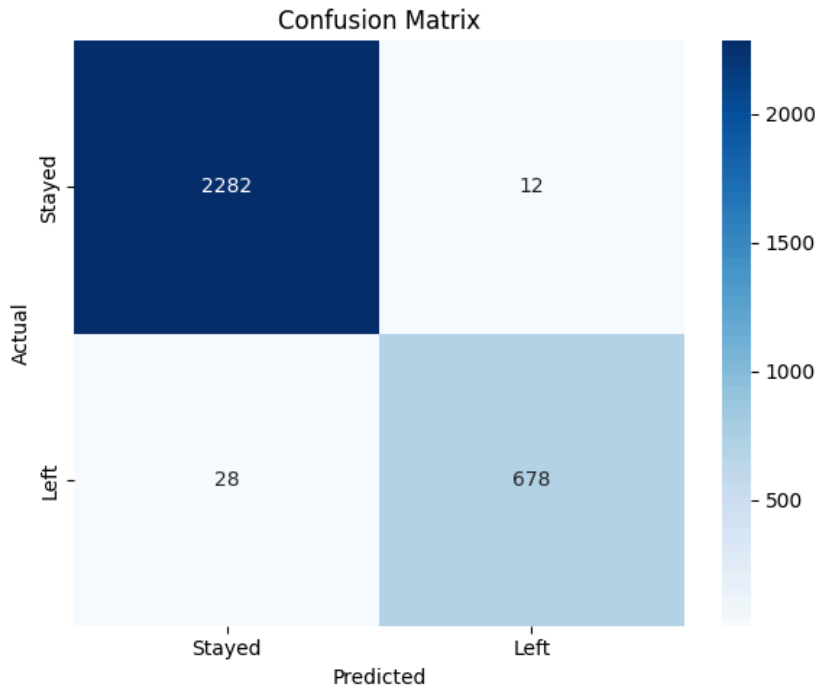
- **Accuracy**: **98.67%** — indicates strong predictive power
- High values across precision, recall, and F1-score suggest reliable model performance

Confusion Matrix:

Code Snippet - Confusion Matrix

```
plt.figure(figsize=(6, 5))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues', xticklabels=["Stayed", "Left"], yticklabels=["Stayed", "Left"])
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.tight_layout()
plt.savefig("Confusion Matrix.png")  # Optional: save
plt.show()
```

Visualization:



The confusion matrix shows that the model makes very few errors when predicting whether employees leave or stay.

Feature Importance:

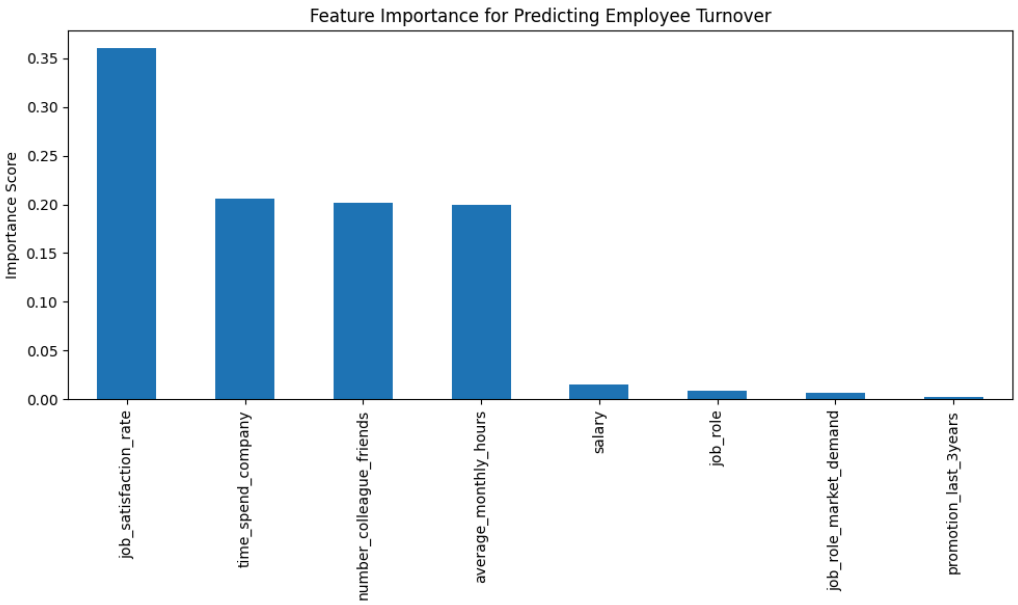
Most Influential Features:

1. Job Satisfaction Rate
2. Time Spent at Company
3. Number of Colleague Friends
4. Average Monthly Hours

Code Snippet - Feature Importance

```
feature_importance = pd.Series(model.feature_importances_, index=X.columns).sort_values(ascending=False)
plt.figure(figsize=(10, 6))
feature_importance.plot(kind='bar')
plt.title("Feature Importance for Predicting Employee Turnover")
plt.ylabel("Importance Score")
plt.tight_layout()
plt.savefig("Feature Importance for Predicting Employee Turnover.png")
plt.show()
```

Visualization:



Less impactful: **Salary, Promotion, Job Role** – indicating emotional and social aspects are more influential than compensation alone.

Insights

Our model suggests employees are more likely to leave due to:

- Low job satisfaction
- Weak workplace relationships
- Prolonged service with limited growth
- Imbalanced workload (too few or too many hours)

References

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The passages in the work, which have been taken from other sources in terms of wording or meaning, are identified by indicating the origin. This also applies to drawings, sketches, picture representations and sources from the Internet.

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Sayeda Fatema Tuj Zohura

Sayeda Fatema Tuj Zohura Erlangen, 2025-07-20